MASTER FUNCTIONAL SKILLS (MFS) · WORKED SOLUTIONS ·
MasterFunctionalSkills.co.uk · AQA Maths Level 2 Specialist
Full Marks (100% Pass) · Verified by Tutor Abdul · January 2020 - Paper 2

Please write clearly in block capitals.

Centre number M F S 0 1 Candidate number 2 0 2 6

Surname MASTER

Forename(s) FUNCTIONAL SKILLS

Candidate signature Master Functional Skills

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Candidate: Master Functional Skills
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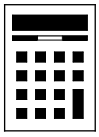
# Functional Skills Level 2 MATHEMATICS

## Paper 2 Calculator

Monday 13 January 2020 Afternoon Time allowed: 1 hour 30 minutes

### Materials

For this paper you must have:
- a calculator
- mathematical instruments.



### Instructions

- Use black ink or black ball-point pen. Draw diagrams in pencil.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- State the units of your answer where appropriate.

| For Examiner's Use | |
|---|---|
| Question | Mark |
| 1–7 | |
| 8 | |
| 9 | |
| 10 | |
| 11 | |
| 12 | |
| **TOTAL** | |

### Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.
- You may ask for more answer paper, graph paper and tracing paper. These must be tagged securely to this answer book.
- If your calculator does not have a π button, take the value of π to be 3.142

### Advice

In all calculations, show clearly how you work out your answer.

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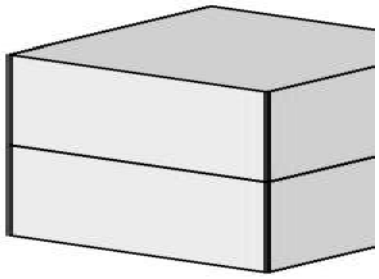
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## Section A

Answer **all** questions in the spaces provided.

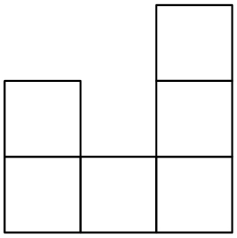
**1** The diagram shows a 3-D shape.



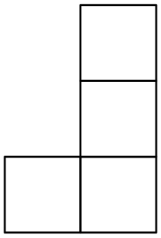
Which is the plan view of the shape?
Circle the correct letter.

**[1 mark]**

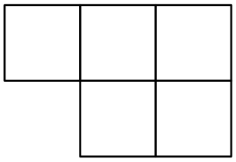
**A**



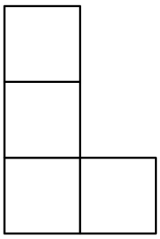
**B**



**C**



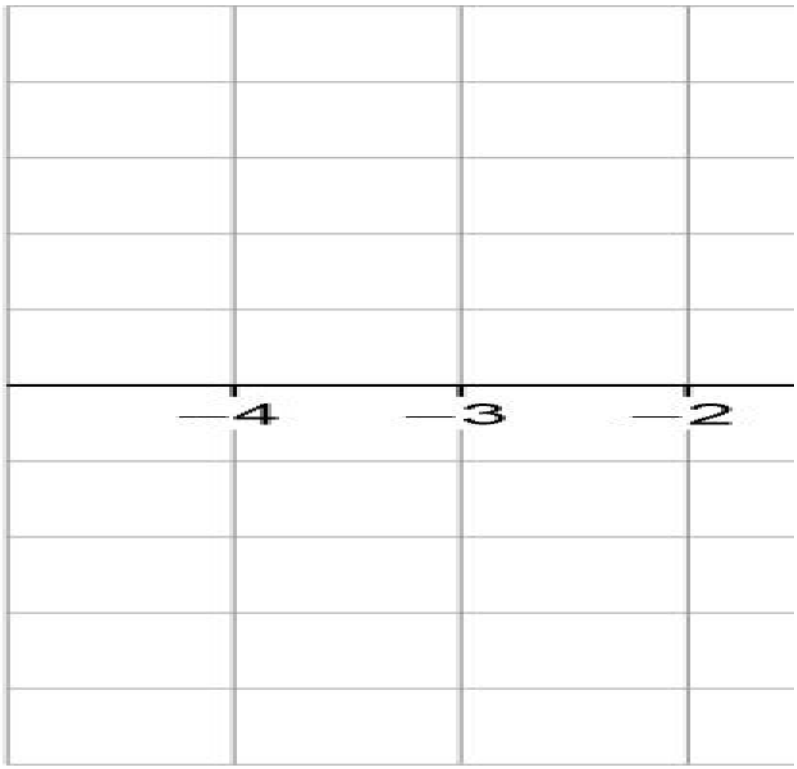
**D**



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**2** Write down the coordinates of the point marked ×.

**[1 mark]**




(x, y)

Answer ( 3 , −2 )

**3** Increase 1670 by 27%

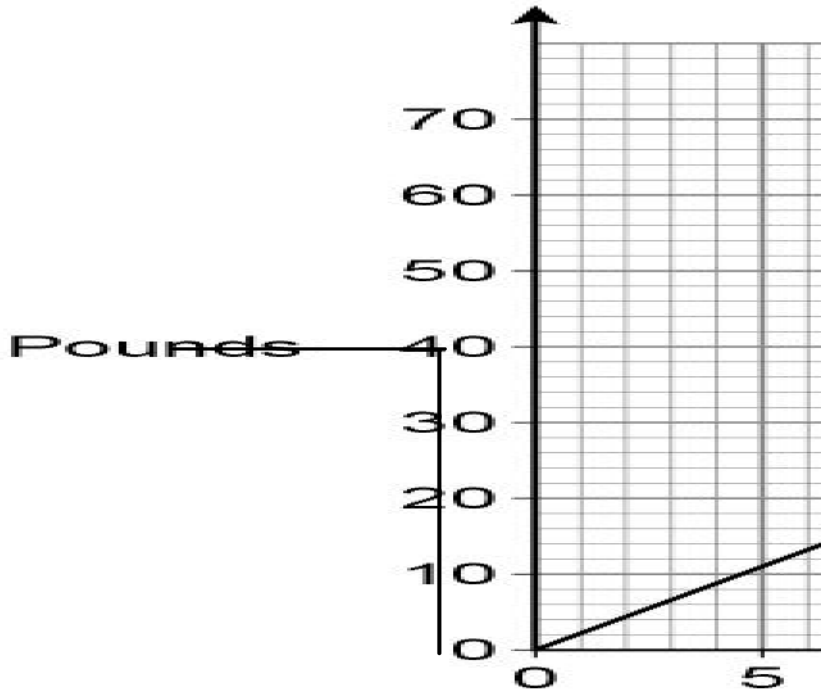
**[2 marks]**

1670 × 1.27 = 2120.9

Answer 2120.9

*Do not write outside the box*

**4** The graph can be used to convert between pounds and kilograms.




Use the graph to convert 40 pounds to kilograms.

**[2 marks]**

Answer 18 kilograms

Do not write outside the box

**5** Calculate $1\frac{3}{4} + 2\frac{4}{5}$

**[1 mark]**

$\frac{7}{3} + \frac{14}{5} = 4\,11/20$

Answer $4\,11/20$

**6** Complete the table to show equivalent fractions, decimals and percentages.

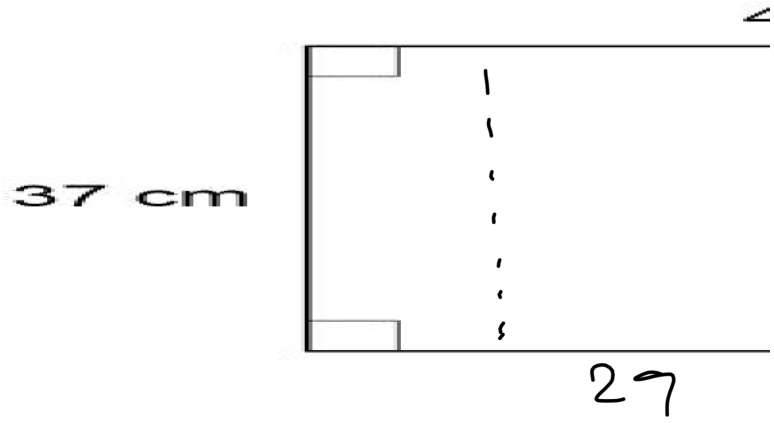
**[2 marks]**

| Fraction | Decimal | Percentage |
|---|---|---|
| $\frac{3}{8}$ | 0.375 | 37.5% |
| $\frac{15}{100}$ | 0.15 | 15% |

**Turn over for the next question**

Do not write outside the box

**7** A shape is made from a rectangle and a triangle.




Not drawn accurately

Work out the area of the shape.

**[3 marks]**

$71 - 42 = 29$

$42 \times 37 = 1554$

$\frac{1}{2} \times 29 \times 37 = 536.5$

$1554 + 536.5 = 2090.5$

Answer 2090.5 $cm^2$

12

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**Section B**

Answer **all** questions in the spaces provided.

**8 Fashion blogger**

Alex is a fashion blogger.

Companies pay her to promote their goods.

**8 (a)** Alex writes 30 posts promoting Company A.

The table shows the number of views of each post.

| Number of views, $x$ | Frequency | mid point | mid x Fq |
|---|---|---|---|
| $0 \leqslant x < 30\,000$ | 5 | 15,000 | 75,000 |
| $30\,000 \leqslant x < 60\,000$ | 9 | 45,000 | 405,000 |
| $60\,000 \leqslant x < 90\,000$ | 12 | 75,000 | 900,000 |
| $90\,000 \leqslant x < 120\,000$ | 4 | 105,000 | 420,000 |
| | Total = 30 | | 1,800,000 |

Alex also writes 23 posts promoting Company B.

The total number of views of these posts is 1 223 600

Which company receives the **higher** average number of views, A or B?

You **must** show your working.

**[4 marks]**

1223,600 ÷ 23 = 53,200

1,800,000 ÷ 30 = 60,000

Company A

**Turn over ►**

Do not write outside the box

Some of the posts were about clothing and the rest were about shoes.

The table shows the number of each.

| | Company A | Company B |
|---|---|---|
| **Clothing** | 21 | 13 |
| **Shoes** | 9 | 10 |

total
34
19
53

One of the posts is chosen at random.

**8 (b)** Work out the probability that it is about clothing.

**[1 mark]**

Answer 34/53

**8 (c)** Work out the probability that it is from Company A **and** is about shoes.

**[1 mark]**

Answer 9/53

Do not write outside the box

**8 (d)** In a year, Alex wrote 1460 posts promoting companies.

She was paid £75 for each post that had at least 65 000 views.

In total, she was paid £36 000

What percentage of her posts had at least 65 000 views?

**[3 marks]**

$36,000 \div 75 = 480$

$\frac{480}{1460} \times 100 = 32.9$

Answer 32.9 %

9

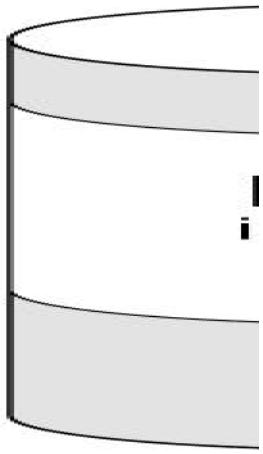
**Turn over for the next question**

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**9 Tinned food**

A company makes tinned food.

**9 (a)** A tin is filled with peas and water.



The contents of the tin have a total mass of 300 grams.

mass of peas : mass of water = 3 : 2

240 of these tins are packed into a box.

Work out the total mass of peas in the box.
Give your answer in kilograms.

**[5 marks]**

3+2 =5 300÷5=60

60x3 =180g of peas per tin

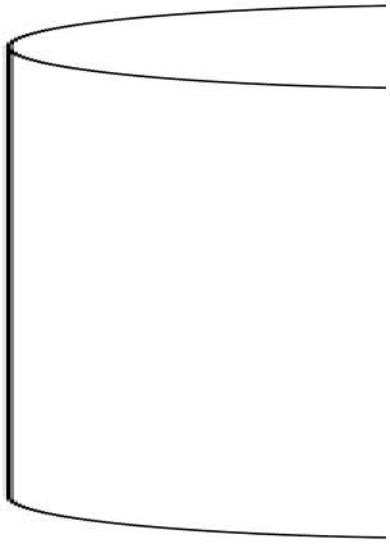
180g =0.18kg

0.18 x240 =43.2

Answer 43.2 kilograms

Do not write outside the box

**9 (b)** A tin for carrots is a cylinder with radius 2.8 cm and height 9 cm



When filled, one sixth of the volume of the tin of carrots is water.

The company uses 425 litres of water each day in tins of carrots.

1 litre = 1000 $cm^3$

How many tins of carrots does the company make each day?

**[5 marks]**

$2.8^2 \times \pi = 24.6\,cm^2$

$24.6 \times 9 = 221.7\,cm^3$

$\frac{1}{6} \times 221.7 = 36.9\,cm^3$ water

$425\,L = 425{,}000\,cm^3$

$425{,}000 \div 36.9 = 11504$

Answer 11504

10

Turn over ►

Do not write outside the box

**10 University student**

Marco is about to start university.

The table shows his total expected income for the **year**.

| | **Amount (£)** |
|---|---|
| **Student loan** | 8700 |
| **Part-time job** | 4800 |

Marco budgets for the following expenses each **month** for 12 months.

| | **Amount per month (£)** |
|---|---|
| **Accommodation** | 589.00 |
| **Living expenses** | 186.00 |
| **Entertainment** | 65.00 |
| **Travel** | 87.50 |

**10 (a)** Marco will save any income that he does not spend.

How much does he expect to save in the **year**?

**[4 marks]**

£700 + £4800 = 13,500

589 + 186 + 65 + 87.50

= 927.50

927.50 × 12 = 11130

13500 − 11130 = 2370

Answer £ 2370

Do not write outside the box

**10 (b)** On Marco's 14th birthday, his parents put £2000 into a bank account for him.
The account pays compound interest at a rate of 1.5% per annum.
Marco can access the account on his 21st birthday.

How much will be in the account on his 21st birthday?

**[3 marks]**

$21 - 14 = 7$

$1.5\% = 0.015$

$1.015^{7} \times 2000 = 2219.69$

Answer £ 2219.69

7

**Turn over for the next question**

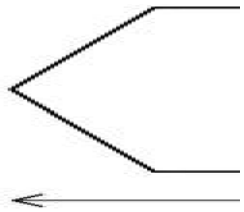
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**11 Jewellery**

**11 (a)** Mo makes and sells jewellery.

He makes pendants in the shape of hexagons.

The hexagons are made of glass and have wire around the perimeter.



Not drawn accurately

Here is a formula for the perimeter, $P$, of the hexagon.

$$P = 6 \times \sqrt{\left(\frac{b}{4}\right)^2 + \left(\frac{w}{2}\right)^2}$$

$b$ is the length of the pendant

$w$ is the width of the pendant

Mo makes pendants with length 6.4 cm and width 5.5 cm

He buys the wire in reels with 4 metres of wire on each reel.

How many pendants can Mo make using one reel of wire?
You **must** show your working.

**[5 marks]**

$\left(\frac{6.4}{4}\right)^2 + \left(\frac{5.5}{2}\right)^2 = 1.6^2 + 2.75^2$

$= 10.1225$

$\sqrt{10.1225} = 3.18$

$6 \times 3.18 = 19.09$

$4m = 400cm$

$400 \div 19.09 = 20.95$

Answer 20

Do not write outside the box

**11 (b)** Mo makes some pendants.

The glass in each pendant is red, blue or yellow.

$\frac{2}{11}$ of the pendants are red

blue pendants and yellow pendants are in the ratio 2 : 1

What fraction of the pendants are blue?

**[4 marks]**

$1 - \frac{2}{11} = \frac{9}{11}$

$2 + 1 = 3$ $\quad \frac{9}{11} \div 3 = \frac{3}{11}$ yellow

$\frac{3}{11} \times 2 = \frac{6}{11}$ blue

Answer $\frac{6}{11}$

9

**Turn over for the next question**

Do not write outside the box

**12 Football match**

Steph supports Leicester City Football Club.

She and three friends are going to watch the team play at West Ham United.

**12 (a)** The table shows information about the three parts of their journey.

| | |
|---|---|
| Drive 108 miles | Average speed 48 miles per hour |
| Train | Maximum 34 minutes |
| Walk to stadium | Maximum 25 minutes |

What is the latest time they can leave to be sure of getting to the stadium by 2.30 pm?
You **must** show your working.

**[4 marks]**

$S = d/t \quad t = d/s$

Drive:

$108/48 = 2.25$ hours

$= 135$ mins

$135 + 34 + 25 = 194$ mins

$= 3$ hours 14 mins

2:30 pm - 3 hours 14 mins

= 11:16 am

Answer 11:16 am

*Do not write outside the box*

**12 (b)** Steph works out the cost of her trip, including the return journey.

She will use her car to drive 108 miles **each way**.

She and her 3 friends will share the cost of fuel equally.

Her car travels 12.5 miles per litre of fuel.

Fuel costs 128.8p per litre.

Steph will also pay

£3 for a return train ticket

£30 for a match ticket

£8 for food.

Work out the total amount Steph will pay for her trip.

**[6 marks]**

$108 \times 2 = 216$

$216 \div 12.5 = 17.28$ litres

$17.28 \times 128.8 = 2225.664$ p

$2225.664 \div 4$ people $= 556.416$p

$556.416\text{p} \approx £5.56$

$5.56 + 3 + 30 + 8 = 46.56$

Answer £ 46.56

**Question 12 continues on the next page**

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**12 (c)** Steph looks at all the match results Leicester City have had at West Ham United.

| Leicester win | West Ham win | Draw |
|---|---|---|
| 12 | 32 | 15 |

Steph says,

"Based on these results, the probability of a Leicester win is more than 20%"

Is she correct?
You **must** show your working.

**[3 marks]**

$12 + 32 + 15 = 59$

$12/59 \times 100 = 20.3\%$

Yes, $20.3\% > 20\%$

13

**END OF QUESTIONS**

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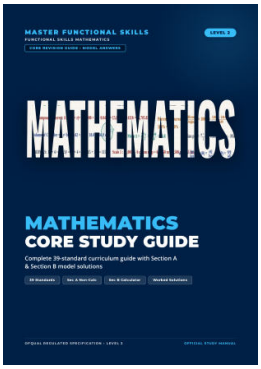
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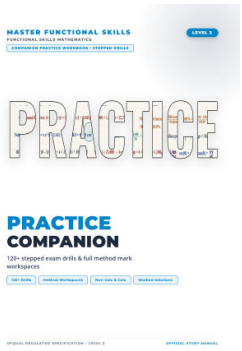



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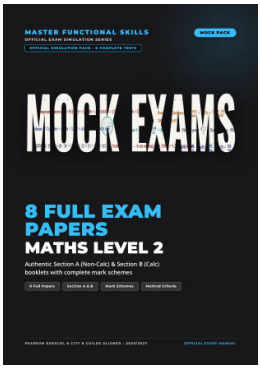



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